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Programming In C-HDOC DAY_13

Problem

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Count the number of even digits, and odd digits. **E.g.** Number: 12045, Even_dig = 3, Odd_dig = 2
- Find the sum of its even digits and the sum of its odd digits separately. **E.g.** Number: 12345, Sum_even = 6, Sum_odd = 9

ALGORITHM

- Start**
- Declare variables c and num.
- Ask the user to enter a positive integer num.
- Ask the user to enter a choice:
 - Choice 1:** Count the number of even and odd digits.
 - Choice 2:** Find the sum of even and odd digits separately.
- Read the choice c.
- Use a switch statement based on c.

If c = 1:

- Initialize even = 0 and odd = 0.
- Repeat while num > 0:
 - Find the last digit using digit = num % 10.
 - If digit % 2 == 0, increment even.
 - Otherwise, increment odd.
 - Remove the last digit using num = num / 10.
- Display the number of even and odd digits.

If c = 2:

- Initialize seven = 0 and sodd = 0.
- Repeat while num > 0:
 - Find the last digit using digit = num % 10.
 - If digit % 2 == 0, add it to seven.
 - Otherwise, add it to sodd.

- Remove the last digit using $\text{num} = \text{num} / 10$.

12. Display the sum of even digits and the sum of odd digits.

If the choice is invalid:

13. Display "Invalid choice".

14. **Stop.**

C PROGRAM CODE

```
#include <stdio.h>
#include <math.h>
int main()
{
    int c;
    int num;
    printf("enter a positive integer number\n");
    scanf("%d", &num);
    printf("choice one is to find the number of even and odd digits in the number entered\n");
    printf("choice 2 is to find the sum of even and odd digits seperately of entered number k\n");
    printf("enter case number\n");
    scanf("%d", &c);
    switch (c)
    {
        case 1:
        {
            int even=0,odd=0;
            int digit;
            while(num>0)
            {
                digit=num%10;
                if(digit%2==0)
                    even++;
                else
                    odd++;
                num=num/10;
            }
        }
```

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```
    printf("no of even digits are %d and odd digits are %d\n",even,odd);
    break;
}
case 2:
{
    int seven=0,sodd=0;
    int digit;
    while(num>0)
    {
        digit=num%10;
        if(digit%2==0)
            seven=seven+digit;
        else
            sodd=sodd+digit;
        num=num/10;
    }
    printf("sum of even digits is %d and sum of odd digits is %d\n",seven,sodd);
    break;
}
default:
    printf("invalid choice\n");
    break;

}
return 0;
}
```

TEST CASES

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	enter a positive integer number 12345678 choice one is to find the number of even and odd digits in the number entered choice 2 is to find the sum of even and odd digits seperately of entered number k enter case number 1	no of even digits are 4 and odd digits are 4	no of even digits are 4 and odd digits are 4	PASS
2	enter a positive integer number 12345 choice one is to find the number of even and odd digits in the number entered choice 2 is to find the sum of even and odd digits seperately of entered number k enter case number 2	sum of even digits is 6 and sum of odd digits is 9	sum of even digits is 6 and sum of odd digits is 9	PASS